Volume Viii Applied Methods Reference

Contents

Calculus

0.1 Limits

Definitions

Computational Methods

0.1.1 Limits

Theorem Reference

Limits Exercises

0.2 Continuity

Definitions

Computational Methods

0.2.1 Continuity

Theorem Reference

Continuity Exercises

0.3 Derivatives

Definition

Derivative Algebra

0.3.1 Derivatives

Standard Derivative Table

Chain-Rule Patterns

Implicit Differentiation

Logarithmic Differentiation

Applications

Derivatives

Derivative Rules

Mixed Derivative Practice

0.4 Integrals

Geometric Modeling

0.7 Curves and Surfaces

0.7.1 Parametric Curves and Surfaces

Parametric Curves

Parametric Surfaces

Partial Derivatives and Isoparametric Curves

Directional Derivative

Unit Normal and Tangent Plane

0.8 Bezier Curves

Proofs

Computational Geometry

0.9 Convex Hulls

Convexity and Hulls

Definition 0.9.1 (Convex Set)**.** A set $S \subseteq \mathbb{R}^2$ is *convex* if for every pair of points $a, b \in S$ and every $\lambda \in [0, 1]$,

$$\lambda a + (1 - \lambda) b \in S.$$

Equivalently, S contains the line segment between any two of its points.

Definition 0.9.2 (Convex Combination)**.** A *convex combination* of points $q_1, \dots, q_n \in \mathbb{R}^2$ is a point of the form

$$\sum_{i=1}^{n} \lambda_i q_i, \qquad \lambda_i \geq 0 \text{ for all } i, \qquad \sum_{i=1}^{n} \lambda_i = 1.$$

The numbers λ_i are called the weights.

Definition 0.9.3 (Convex Hull)**.** The *convex hull* of a set $P \subseteq \mathbb{R}^2$, written $\operatorname{conv}(P)$, is the set of all convex combinations of finitely many points of P:

$$\operatorname{conv}(P) = \left\{ \sum_{i=1}^{n} \lambda_i q_i \;\middle|\; n \geq 1,\ q_i \in P,\ \lambda_i \geq 0,\ \sum_{i=1}^{n} \lambda_i = 1 \right\}.$$

Equivalently, $\operatorname{conv}(P)$ is the smallest convex set containing P, i.e. the intersection of all convex sets that contain P.

Lemma 0.9.4 (Convex hull as the smallest convex set)**.** *Let $P \subseteq \mathbb{R}^2$. Then $\operatorname{conv}(P)$ is convex, contains P, and is contained in every convex set that contains P. Hence*

$$\operatorname{conv}(P) = \bigcap \{ K \subseteq \mathbb{R}^2 : K \textit{ is convex and } P \subseteq K \}.$$

Supporting Functionals

Definition 0.9.5 (Extreme Point)**.** Let $S \subseteq \mathbb{R}^2$ be convex. A point $p \in S$ is an *extreme point* of S if whenever

$$p = \lambda a + (1 - \lambda) b, \qquad a, b \in S, \quad \lambda \in (0, 1),$$

it follows that $a = b = p$. For a finite point set P, the extreme points of $\operatorname{conv}(P)$ are called the vertices of the convex hull.

Definition 0.9.6 (Supporting Line)**.** A line $\ell \subseteq \mathbb{R}^2$ is a *supporting line* of a convex set S at a point $p \in S$ if $p \in \ell$ and S lies entirely in one of the two closed half-planes bounded by ℓ. Equivalently, there exists $d \in \mathbb{R}^2 \setminus \{0\}$ such that

$$\ell = \{ x \in \mathbb{R}^2 : \langle d, x \rangle = \langle d, p \rangle \}$$

and

$$\langle d, q \rangle \leq \langle d, p \rangle \qquad \text{for all } q \in S.$$

Lemma 0.9.7 (Linear functionals preserve convex combinations)**.** *Let $d \in \mathbb{R}^2$, and let*

$$z = \sum_{i=1}^{n} \lambda_i q_i$$

be a convex combination of points $q_1, \ldots, q_n \in \mathbb{R}^2$. Then

$$\langle d, z \rangle = \sum_{i=1}^{n} \lambda_i \langle d, q_i \rangle.$$

Consequently, if $\langle d, q_i \rangle \le M$ for every i, then

$$\langle d, z \rangle \le M.$$

Theorem 0.9.8 (Supporting functional characterization of hull vertices)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. A point $p \in P$ is a vertex of $\mathrm{conv}(P)$ if and only if there exists a direction $d \in \mathbb{R}^2 \setminus \{0\}$ such that*

$$\langle d, p \rangle > \langle d, q \rangle \qquad \text{for all } q \in P \setminus \{p\}.$$

Equivalently, p is the unique point of P attaining

$$\max_{q \in P} \langle d, q \rangle.$$

Thus every hull vertex is a maximum of P under some linear functional.

Upper Hull Structure

Theorem 0.9.9 (Monotone slope characterization of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Let*

$$v_0, v_1, \ldots, v_m$$

be the vertices of the upper hull of P, listed in increasing x-coordinate, and define

$$s_k = \frac{y(v_{k+1}) - y(v_k)}{x(v_{k+1}) - x(v_k)}, \qquad k = 0, 1, \ldots, m-1.$$

Then

$$s_0 > s_1 > \cdots > s_{m-1}.$$

Conversely, if points $w_0, \ldots, w_m$ satisfy

$$x(w_0) < x(w_1) < \cdots < x(w_m)$$

and their consecutive slopes are strictly decreasing, then every w_k is a vertex of

$$\mathrm{conv}(\{w_0, \ldots, w_m\}).$$

The lower hull is characterized similarly, with strictly increasing consecutive slopes.

Theorem 0.9.10 (Greedy slope construction of the upper hull)**.** *Let $P \subset \mathbb{R}^2$ be finite and in general position. Define*

$$v_0 = \arg\min_{p \in P} x(p).$$

Given v_k with

$$x(v_k) < \max_{p \in P} x(p),$$

define

$$v_{k+1} = \arg\max_{\substack{q \in P \\ x(q) > x(v_k)}} \frac{y(q) - y(v_k)}{x(q) - x(v_k)}.$$

Terminate once

$$v_k = \arg\max_{p \in P} x(p).$$

Then the points

$$v_0, v_1, \ldots, v_m$$

produced by this construction are exactly the vertices of the upper hull of P, and the slopes encountered along the way are strictly decreasing.

0.10 Convex Hulls

Proofs

Computational Linear Algebra

0.11 Vectors

0.11.1 Vectors and Linear Combinations

Vectors

Linear Combinations

0.11.2 Lengths and Dot Products

The Dot Product

Length and Angle

0.11.3 Introduction to Matrices

Matrices and Matrix–Vector Multiplication

The Transpose

Matrix Form of a Curve

0.12 Solving Linear Equations

0.12.1 Rules for Matrix Operations

Algebraic Rules

Diagonal Matrices

Block Matrices

0.12.2 Inverse Matrices

Definition and Properties

The 2×2 Case

Change of Basis via Inverse

0.13 Vector Spaces

0.13.1 Independence, Basis, and Dimension

Linear Independence

Basis and Dimension

Change of Basis

Fourier Analysis

Proofs

Ordinary Differential Equations

0.23 Breadcrumb

0.24 Models

Definition 0.24.1 (Differential Equation)**.** A **differential equation** is an equation whose unknown is a function and whose terms involve one or more derivatives of that function.
For an unknown function $y = y(t)$, a differential equation may be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0.$$

Here t is the independent variable, $y(t)$ is the unknown function, and

$$y', y'', \ldots, y^{(n)}$$

are derivatives of y.

Definition 0.24.2 (Order of a Differential Equation)**.** The **order** of a differential equation is the highest derivative of the unknown function that appears in the equation.
For example, if an equation involving an unknown function $y = y(t)$ can be written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0$$

and $y^{(n)}$ is the highest derivative that appears, then the differential equation has **order** n.

Definition 0.24.3 (Solution of a Differential Equation)**.** A **solution** of a differential equation is a function that satisfies the equation on a specified interval.
More precisely, if a differential equation is written in the form

$$F(t, y, y', y'', \ldots, y^{(n)}) = 0,$$

then a solution on an interval I is a function $y : I \to \mathbb{R}$ such that y has all derivatives required by the equation and

$$F(t, y(t), y'(t), y''(t), \ldots, y^{(n)}(t)) = 0$$

for every $t \in I$.

Definition 0.24.4 (Autonomous Differential Equation)**.** An **autonomous differential equation** is a differential equation in which the independent variable does not appear explicitly.
For a first-order ordinary differential equation, an autonomous equation has the form

$$y' = f(y),$$

where the rate of change of y depends only on the current value of y, not directly on time t.

Definition 0.24.5 (Non-Autonomous Differential Equation)**.** A **non-autonomous differential equation** is a differential equation in which the independent variable appears explicitly.

For a first-order ordinary differential equation, a non-autonomous equation has the form

$$y' = f(t, y),$$

where the rate of change of y may depend both on the current value of y and on the independent variable t.

Definition 0.24.6 (Linear Differential Equation)**.** A differential equation is called **linear** if the unknown function and its derivatives appear only to the first power and are not multiplied together.

An nth-order linear ordinary differential equation for an unknown function $y = y(t)$ has the form

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

where the coefficient functions

$$a_0(t), a_1(t), \ldots, a_n(t)$$

and the forcing term $g(t)$ are given functions of the independent variable t.

Definition 0.24.7 (Malthusian Population Growth Model)**.** The **Malthusian population growth model** assumes that the rate of change of a population is proportional to the current population.

If $P(t)$ denotes the population at time t, then the model is

$$P'(t) = kP(t),$$

where k is a constant called the **growth rate** or **proportionality constant**.

Definition 0.24.8 (Decay)**.** A quantity is said to undergo **decay** if it decreases over time.

In a differential equation model, if $Q(t)$ denotes the quantity at time t, then decay is often modeled by an equation of the form

$$Q'(t) = -kQ(t),$$

where $k > 0$ is a constant.

Definition 0.24.9 (Newton's Law of Cooling)**.** **Newton's Law of Cooling** states that the rate at which an object's temperature changes is proportional to the difference between the object's temperature and the surrounding ambient temperature.

If $T(t)$ denotes the temperature of the object at time t, and T_a denotes the ambient temperature, then the model is

$$T'(t) = -k\big(T(t) - T_a\big),$$

where $k > 0$ is a constant called the **cooling constant**.

Definition 0.24.10 (General Solution)**.** A **general solution** of a differential equation is a family of solutions containing one or more arbitrary constants.

For an nth-order ordinary differential equation, the general solution typically contains n arbitrary constants:

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n).$$

Each choice of the constants $C_1, C_2, \ldots, C_n$ gives a particular solution of the differential equation.

Definition 0.24.11 (Family of Solutions)**.** A **family of solutions** of a differential equation is a collection of solution functions described by one or more parameters.

For example, a family of solutions may be written in the form

$$y(t) = \Phi(t, C),$$

or more generally,

$$y(t) = \Phi(t, C_1, C_2, \ldots, C_n),$$

where $C, C_1, \ldots, C_n$ are parameters. Each choice of the parameters gives one particular solution.

Definition 0.24.12 (Initial Condition)**.** An **initial condition** specifies the value of the unknown function at a particular value of the independent variable.

For a first-order differential equation involving an unknown function $y = y(t)$, an initial condition has the form

$$y(t_0) = y_0,$$

where t_0 is the initial time and y_0 is the initial value.

Definition 0.24.13 (Initial Value Problem)**.** An **initial value problem**, or **IVP**, is a differential equation together with one or more initial conditions.

For a first-order differential equation, an initial value problem has the form

$$y' = f(t, y), \qquad y(t_0) = y_0.$$

The differential equation describes how the function changes, while the initial condition specifies the value of the solution at the initial time t_0.

Definition 0.24.14 (Existence)**.** An initial value problem is said to have **existence** if there is at least one function that satisfies both the differential equation and the initial condition.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

existence means that there is a function $y(t)$, defined on some interval containing t_0, such that

$$y'(t) = f(t, y(t))$$

and

$$y(t_0) = y_0.$$

Definition 0.24.15 (Uniqueness)**.** An initial value problem is said to have **uniqueness** if it has at most one solution on the interval under consideration.

For a first-order initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

uniqueness means that if $y_1(t)$ and $y_2(t)$ are both solutions on the same interval I containing t_0, then

$$y_1(t) = y_2(t)$$

for every $t \in I$.

Definition 0.24.16 (Interval of Existence)**.** An **interval of existence** for a solution of an initial value problem is an interval on which the solution is defined and satisfies the differential equation.

For an initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0,$$

an interval I is an interval of existence if $t_0 \in I$ and there is a solution $y : I \to \mathbb{R}$ such that

$$y'(t) = f(t, y(t))$$

for every $t \in I$, and

$$y(t_0) = y_0.$$

Proofs

Partial Differential Equations

0.25 Partial Differential Equations

Proofs

Computational Probability

0.26 Computational Probability

Proofs